

THE SAILING LEAGUE PROBLEM AT TEN TEAMS AND TWENTY-FOUR FLIGHTS: THE CASE $n = 3$, $\text{opt}(10, 24, 5) = 2$

ABSTRACT. Schüler and Schürmann introduced the *sailing league problem*: for parameters $(N_{\text{teams}}, N_{\text{flights}}, N_{\text{inrace}})$, minimise $\lambda_{\max} - \lambda_{\min}$ over pairing lists. At the end of their case study of the Asian Pacific Sailing Champions League they record that the case $N_{\text{teams}} = 10$, $N_{\text{inrace}} = 5$, $N_{\text{flights}} = 8n$ with $n \geq 3$ “still appears to be open”. We settle exactly one member of that family, the case $n = 3$: writing $\text{opt}(N_{\text{teams}}, N_{\text{flights}}, N_{\text{inrace}})$ for the minimum,

$$\text{opt}(10, 24, 5) = 2.$$

The upper bound is an explicit 24×10 tournament plan, printed here in full, whose 45 pair counts are 30 tens and 15 twelves; it is a pairing list in the source’s sense and repeats three of its flights. The lower bound uses no computer: $\lambda = 32/3$ is not an integer, and utility 1 would force a 6-regular graph on 10 vertices in which every triple spans an even number of edges, which does not exist. The cases $n \geq 4$ are not settled here.

1. THE RESULT

Fix $N_{\text{teams}} = 10$ and $N_{\text{inrace}} = 5$ and write a *tournament plan* on F flights as an $F \times 10$ matrix M over $\{1, 2\}$: the entry $M[f][t]$ names which of the two races of flight f team t sails in, and every row must carry exactly five 1s and five 2s, since each flight partitions the ten teams into two races of five. For teams $t \neq u$ put

$$\lambda(t, u) = \#\{f : M[f][t] = M[f][u]\},$$

the number of races in which t and u meet, and let the *utility value* be $\lambda_{\max} - \lambda_{\min}$ over the 45 unordered pairs. The sailing league problem asks to minimise it; write $\text{opt}(N_{\text{teams}}, N_{\text{flights}}, N_{\text{inrace}})$ for the minimum.

Theorem 1. $\text{opt}(10, 24, 5) = 2$.

The upper bound is the plan of Table 1 (Section 3); the matching lower bound is Theorem 5 in Section 4. Both halves are checkable by hand from objects printed below.

2. THE OPEN STATEMENT, VERBATIM

All quotations are from the arXiv e-print L^AT_EX source of [6], in which the macros `\Nteams`, `\Nflights`, `\Ninrace` expand to N_{teams} , N_{flights} , N_{inrace} .

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The open statement is unnumbered prose at the end of Section 5.1 (“Asian Pacific Sailing Champions League”), immediately before Section 5.2:

We finally note that the problem of finding an optimal tournament plan for parameters $N_{\text{teams}} = 10$, $N_{\text{inrace}} = 5$ and $N_{\text{flights}} = n \cdot 8$ with $n \geq 3$ (n -times the base schedule of the Asian Pacific Champions League) still appears to be open.

The quantifier is as printed: for all integers $n \geq 3$. This note settles $n = 3$, i.e. $N_{\text{flights}} = 24$, and nothing beyond it.

The parenthetical is a gloss on the *flight count*, not a structural demand that the plan consist of n copies of one 8-flight base schedule. The sentence names the parameter triple, and the source’s own definition of the problem (line 252) is stated for the triple:

For parameters N_{teams} , N_{flights} , N_{inrace} the corresponding *sailing league problem* asks to find a pairing list with $\lambda_{\max} - \lambda_{\min}$ as small as possible.

The authors’ own $n = 2$ answer is likewise not a repetition: in their 16-flight table, rows 9–16 are not a repeat of rows 1–8; only rows 1–8 double as the 8-flight optimum.

The definition the result depends on is at lines 192–203: a *pairing list* is a set T of N_{teams} teams, a list R of races each of cardinality N_{inrace} , and a list F of N_{flights} flights partitioning R such that each flight partitions T . Since $N_{\text{inrace}} = 5$ divides $N_{\text{teams}} = 10$ with quotient 2, each flight is exactly a 5/5 split, which is the matrix formulation of Section 1. Their averaging lemma (line 235) gives

$$(1) \quad \lambda = \frac{N_{\text{flights}} (N_{\text{inrace}} - 1)}{N_{\text{teams}} - 1} = \frac{24 \cdot 4}{9} = \frac{96}{9} = \frac{32}{3},$$

which is not an integer.

The two solved members are Theorems at lines 1015–1022 and 1052–1057:

For the parameters $N_{\text{teams}} = 10$, $N_{\text{flights}} = 8$ and $N_{\text{inrace}} = 5$ an optimal tournament plan has a utility value $\lambda_{\max} - \lambda_{\min} = 3$ and satisfies $\lambda_{\min} = 2$, $\lambda_{\max} = 5$.

For the parameters $N_{\text{teams}} = 10$, $N_{\text{flights}} = 16$ and $N_{\text{inrace}} = 5$ an optimal tournament plan has a utility value $\lambda_{\max} - \lambda_{\min} = 2$.

The value at 24 flights is not settled in the cited source: the string $N_{\text{flights}} = 24$ does not occur in its L^AT_EX file, and its appendix of best-known plans contains only a (32, 18, 8) plan and the (10, 16, 5) plan.

3. THE EXHIBITED PLAN

Table 1 is a (10, 24, 5) tournament plan. A referee needs no program: check that each row carries five 1s and five 2s, then count same-symbol columns for each of the 45 pairs.

The canonical machine-readable form of the plan is the following list of 24 row strings, flight 1 first, character-exact against Table 1.

THE SAILING LEAGUE PROBLEM AT TEN TEAMS AND TWENTY-FOUR FLIGHTS3

TABLE 1. A tournament plan for $N_{\text{teams}} = 10$, $N_{\text{flights}} = 24$,
 $N_{\text{inrace}} = 5$ with $\lambda_{\text{max}} - \lambda_{\text{min}} = 2$.

Flight		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
1		1	1	1	1	2	2	2	2	1	2
2		1	1	1	2	1	2	1	2	2	2
3		1	1	1	2	2	1	2	1	2	2
4		1	1	2	1	1	1	2	2	2	2
5		1	2	2	2	2	2	1	1	1	1
6		1	2	1	1	1	1	2	2	2	2
7		1	1	2	1	2	2	1	2	2	1
8		1	2	2	1	2	1	1	2	1	2
9		1	2	1	2	2	1	2	2	1	1
10		1	1	2	1	2	2	1	2	2	1
11		1	2	2	1	2	1	1	2	1	2
12		1	2	1	2	2	1	2	2	1	1
13		1	1	2	1	2	2	2	1	1	2
14		1	2	2	1	2	1	2	1	2	1
15		1	2	1	2	2	1	1	1	2	2
16		1	1	2	2	1	1	1	2	2	2
17		1	2	2	2	1	2	1	2	1	1
18		1	2	1	1	1	2	2	2	1	2
19		1	1	2	2	1	2	2	1	1	2
20		1	2	2	1	1	2	2	1	2	1
21		1	2	1	1	2	2	1	1	2	2
22		1	1	2	2	2	1	2	1	2	1
23		1	2	2	2	1	1	1	1	2	2
24		1	2	1	2	1	2	1	2	2	1

1111222212 1112121222 1112212122 1121112222
1222221111 1211112222 1121221221 1221211212
1212212211 1121221221 1221211212 1212212211
1121222112 1221212121 1212211122 1122111222
1222121211 1211122212 1122122112 1221122121
1211221122 1122212121 1222111122 1212121221

The complete table of pair counts (teams 1 to 10, symmetric, diagonal blank) is

t= 1:	.	10	10	12	10	12	12	10	10	10
t= 2:	10	.	10	12	12	10	10	12	10	10
t= 3:	10	10	.	10	12	12	10	10	12	10
t= 4:	12	12	10	.	10	10	10	10	12	10
t= 5:	10	12	12	10	.	10	12	10	10	10
t= 6:	12	10	12	10	10	.	10	12	10	10
t= 7:	12	10	10	10	12	10	.	10	10	12

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t= 8:  10  12  10  10  10  12  10   .  10  12
t= 9:  10  10  12  12  10  10  10  10   .  12
t=10:  10  10  10  10  10  10  12  12  12   .

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so $\lambda_{\min} = 10$, $\lambda_{\max} = 12$, and the utility is 2; the profile is exactly $\{10 : 30, 12 : 15\}$, and no pair sits at 11.

Remark (three repeated flights). The plan uses 21 distinct flights, not 24: flight 7 equals flight 10, flight 8 equals flight 11, and flight 9 equals flight 12. This is legal by the source’s own definition, one line after the pairing-list definition (line 205):

Note that we use the term `\emph{list}` to emphasize that it is allowed to repeat races and even flights.

and line 206: “Mathematically, one could also use an *incidence structure* or a *multiset* here.” So a pairing list is a multiset of flights, and Table 1 is a legal $(10, 24, 5)$ pairing list, hence a feasible point of exactly the problem the open sentence poses. The authors’ published 16-flight optimum uses 16 distinct flights. The distinct-flights variant — is there a 24-flight $(10, 5)$ plan of utility 2 with 24 pairwise distinct flights? — is a different question, which this note does not answer in either direction.

4. THE LOWER BOUND

No computer is needed for any step of this section.

Lemma 2 (the authors’ average at this cell). *In each flight a team sails against exactly $N_{\text{inrace}} - 1 = 4$ opponents, so $\sum_{u \neq t} \lambda(t, u) = 24 \cdot 4 = 96$ for every team t , and the grand total over the 45 pairs is $10 \cdot 96/2 = 480$. Consequently $\lambda = 96/9 \notin \mathbb{Z}$ and no $(10, 24, 5)$ pairing list has utility 0.*

Proof. Utility 0 means $\lambda_{\min} = \lambda_{\max}$, and then that common value equals the average λ , which must therefore be an integer. This half is the authors’: their lemma at lines 389–392 states that a perfect pairing list forces $\lambda \in \mathbb{Z}$, equivalently $(N_{\text{teams}} - 1) \mid N_{\text{flights}}(N_{\text{inrace}} - 1)$. Here $9 \nmid 96$. $\square$

Lemma 3 (triangle parity). *For any $(10, F, 5)$ pairing list and any three distinct teams t, u, v ,*

$$\lambda(t, u) + \lambda(t, v) + \lambda(u, v) \equiv F \pmod{2}.$$

In particular at $F = 24$ every triangle sum is even.

Proof. Fix one flight. It splits the ten teams into two races of five, so the three teams fall $3 + 0$ or $2 + 1$ across the two races. In the first case all three pairs are same-race, contributing 3; in the second exactly one pair is, contributing 1. Either way the contribution is odd, so the total over F flights is congruent to F modulo 2. $\square$

The proof uses only that each flight has exactly *two* races, so Lemma 3 does not transfer to the 18-team or 32-team case studies of [6] and is claimed for none of them.

Lemma 4 (the odd graph is a cut). *Let G be a graph on the ten teams all of whose 120 triples span an even number of edges. Then G is a complete bipartite cut: there is $U \subseteq T$ with $E(G) = \{tu : |\{t, u\} \cap U| = 1\}$. Its degrees are $10 - |U|$ on U and $|U|$ off U , and its edge count $|U|(10 - |U|)$ lies in $\{0, 9, 16, 21, 24, 25\}$.*

Proof. Write $p(t, u) \in \{0, 1\}$ for the indicator of $tu \in E(G)$. Fix a team t_0 , set $f(t_0) = 0$ and $f(u) = p(t_0, u)$ for $u \neq t_0$. For $u, v \neq t_0$ the triple (t_0, u, v) has even p -sum, so $p(u, v) = f(u) + f(v) \bmod 2$; for $u \neq t_0$ this also holds by construction. Hence $tu \in E(G)$ exactly when $f(t) \neq f(u)$, i.e. G is the cut determined by $U = f^{-1}(1)$. The degree and edge-count statements are immediate, and $k(10 - k)$ for $k = 0, \dots, 5$ gives $0, 9, 16, 21, 24, 25$. $\square$

Theorem 5. $\text{opt}(10, 24, 5) \geq 2$.

Proof. Utility 0 is impossible by Lemma 2. Suppose some plan has utility 1, so all 45 values lie in a window $\{m, m + 1\}$. By Lemma 2 the per-team sum is 96 over nine neighbours, so $9m \leq 96 \leq 9m + 9$, i.e. $9.67 \leq m \leq 10.67$, forcing $m = 10$ and the window $\{10, 11\}$. Let $a(t) = \#\{u : \lambda(t, u) = 11\}$; then $10 \cdot 9 + a(t) = 96$, so $a(t) = 6$ for every team and the graph of pairs at 11 is 6-regular, with 30 edges. But 11 is the only odd value in the window, so that graph is precisely the graph of pairs with odd λ , which by Lemmas 3 and 4 is a cut $K_{k, 10-k}$. Six-regularity would need $10 - k = 6$ and $k = 6$ at once, i.e. $k = 4$ and $k = 6$: a contradiction. $\square$

Theorem 5 and Table 1 together prove Theorem 1.

5. ATTRIBUTION, AND WHAT IS NOT SETTLED

Due to Schüler and Schürmann [6]. The problem, all definitions, and the open statement settled here at $n = 3$; the kill of utility 0, which is their lemma at lines 389–392; and the theorems $\text{opt}(10, 8, 5) = 3$ and $\text{opt}(10, 16, 5) = 2$, quoted in Section 2.

Not settled in the cited source. The plan of Table 1 and the parity route of Lemmas 3 and 4 and Theorem 5 do not occur in [6], which contains no parity argument and leaves the value at 24 flights open. For the wider literature see “Adjacent work” below.

Not settled.

- *Only the case $n = 3$ is settled.* The source sentence quantifies over all $n \geq 3$. Nothing here settles $n = 4$ (32 flights) or any larger n , and no witness at any flight count other than 24 appears in this note.
- *The plan repeats three flights*, which the source’s definition permits. The distinct-flights variant of this cell, and its optimum, are not addressed.

Adjacent work. In the language of designs, the data of Table 1 are $v = 10$ points, blocks of size $k = 5$ resolved into the 24 flights, replication $r = 24$, and the two concurrences 10 and 12. A published line minimises exactly

this spread of concurrences in a resolvable design — the bridge-club designs, equivalently resolvable regular graph designs — in cases of block size 4: Elenbogen and Maxim [2] ask for 12 players in three tables of four over eight meetings with each pair meeting two or three times and report an unsuccessful search; Royle and Wallis [4] construct such a design and find exactly two isomorphism classes; Kreher, Royle and Wallis [3] treat resolvable regular graph designs of block size 4 on 8, 12 and 16 points, the 8-point case being the two-blocks-per-class analogue of the cell considered here; see also Wallis [7, 8]. None of these treats $v = 10$ with $k = 5$, or $r = 24$, so none of them settles $\text{opt}(10, 24, 5)$. Of the two-associate-class catalogues, Saurabh [5] describes its own scope as the classical window $2 \leq r, k \leq 10$ used by Clatworthy [1] and tabulates designs on 15 and 21 points; $r = 24$ lies outside that window. Neither catalogue was consulted page by page — both are paywalled — and Google Scholar and MathSciNet were not reachable when this was written; citer registries for [6] report no citing work, and keyword sweeps return only the source itself. No priority claim is made.

6. REPRODUCTION AND VERIFICATION

Every input needed to redo the work is printed above, and a reader who trusts no program can verify Theorem 1 by hand.

The plan of Table 1 was found by a constraint solver. That is provenance only: the captured output of the search is not part of this record and no program re-runs it, since an object a solver returns is checked directly.

The checks were also redone in exact arithmetic with no third-party package: the plan re-read from both of its printed forms and its 45 pair counts, profile, totals and repeated flights re-derived; Lemma 3 verified on all 126 five-five splits against all 120 triples; and Lemma 4 verified by exhausting every graph on ten labelled vertices all of whose triples span an even number of edges (there are exactly 512, each of them a cut, none 6-regular). The program and a transcript of it being run accompany this note; neither is needed in order to redo the work, and neither checks any bibliographic or prior-art statement — those are reads, and their limits are recorded in Section 5.

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